

PRAKTIKUM SISTEM OPERASI

MODUL 3
EKSPLORASI XINU



Oleh:

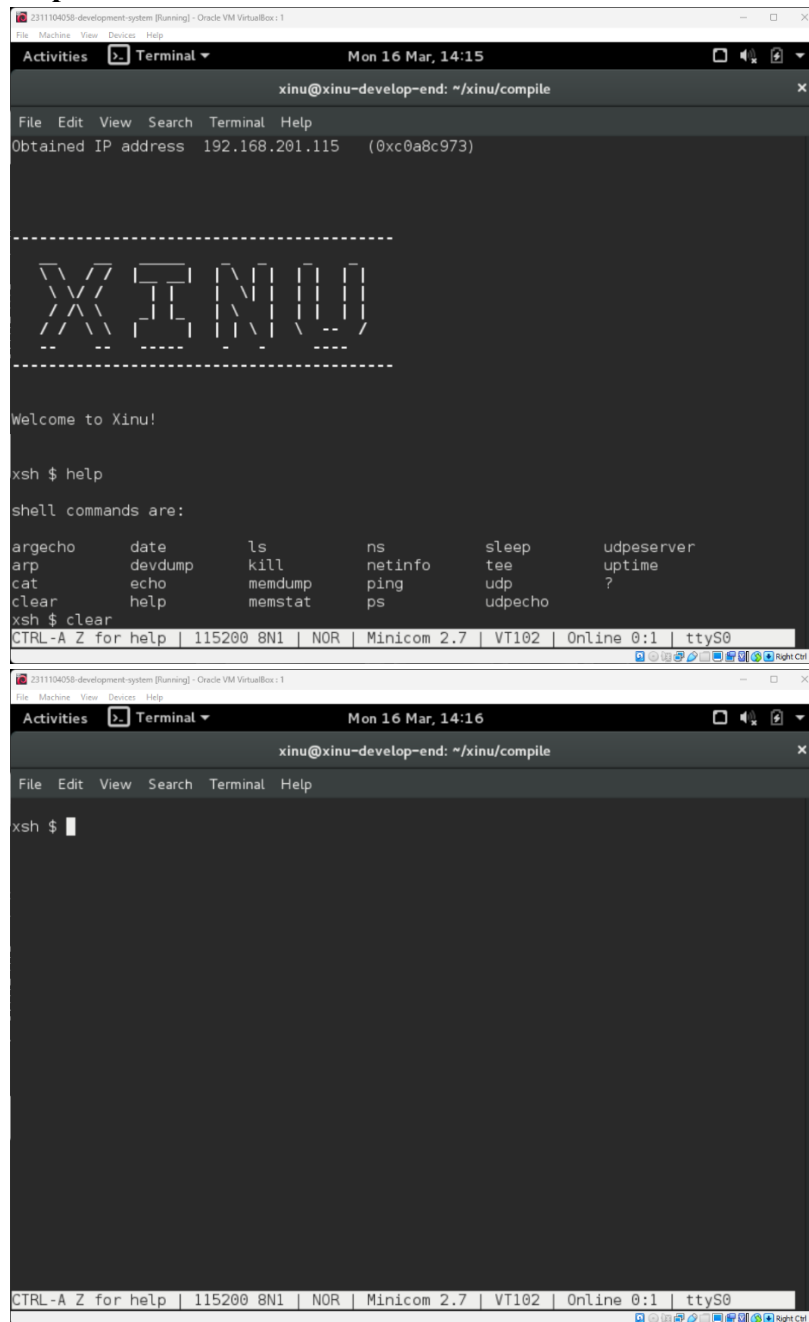
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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK
DIREKTORAT KAMPUS PURWOKERTO
UNIVERSITAS TELKOM
2026

JURNAL

1. help Xinu OS



The image displays two screenshots of a terminal window running Xinu OS. The top screenshot shows the boot process, including the IP address 192.168.201.115 and the ASCII art logo for XINU. The bottom screenshot shows the help command output, listing various shell commands.

```
2311104058-development-system [Running] - Oracle VM VirtualBox : 1
File Machine View Devices Help
Activities Terminal Mon 16 Mar, 14:15
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help
Obtained IP address 192.168.201.115 (0xc0a8c973)

-----
X I N U
-----

Welcome to Xinu!

xsh $ help

shell commands are:

argecho    date      ls        ns        sleep     udpserver
arp        devdump  kill      netinfo   tee       uptime
cat        echo     memdump  ping      udp       ?
clear     help    memstat  ps        udpecho

xsh $ clear
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:1 | ttyS0
```

```
2311104058-development-system [Running] - Oracle VM VirtualBox : 1
File Machine View Devices Help
Activities Terminal Mon 16 Mar, 14:16
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

xsh $
```

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:1 | ttyS0

```
2311104050-development-system [Running] - Oracle VM VirtualBox: 1
File Machine View Devices Help
Activities Terminal Mon 16 Mar, 14:17
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
xsh $ echo Howdy World
Howdy World
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:2 | ttyS0
```

```
2311104050-development-system [Running] - Oracle VM VirtualBox: 1
File Machine View Devices Help
Activities Terminal Mon 16 Mar, 14:18
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
xsh $ uptime
Xinu has been up 3 minute(s) 24 second(s)
xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:3 | ttyS0
```

```
2311104050-development-system [Running] - Oracle VM VirtualBox: 1
File Machine View Devices Help
Activities Terminal Mon 16 Mar, 14:23
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

xsh $ help
shell commands are:
argecho    date      ls         ns         sleep      udpserver
arp        devdump  kill       netinfo    tee         uptime
cat        echo      memdump    ping       udp         ?
clear      help     memstat    ps         udpecho
xsh $ argecho

The 1 arguments are:
0: argecho

xsh $ arp
ARP cache:
  State Pid   IP Address   Hardware Address
  -----
  RESLV   192.168.201. 1 08:00:27:41:25:10

xsh $ cat
miaw
miaw
^Dxsh $

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:9 | ttyS0
```

```
2311104050-development-system [Running] - Oracle VM VirtualBox: 1
File Machine View Devices Help
Activities Terminal Mon 16 Mar, 14:23
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

xsh $ devdump
Device      Name      Minor
-----
0  CONSOLE  0
1  NULLDEV  0
2  ETHER0   0
3  NAMESPACE 0
4  RDISK    0
5  RAM0     0
6  RFILESYS 0
7  RFILE0   0
8  RFILE1   1
9  RFILE2   2
10 RFILE3   3
11 RFILE4   4
12 RFILE5   5
13 RFILE6   6
14 RFILE7   7
15 RFILE8   8
16 RFILE9   9
17 LFILESYS 0
18 LFILE0   0
19 LFILE1   1
20 LFILE2   2
21 LFILE3   3
22 LFILE4   4
23 LFILE5   5
24 PIPE     0
```

```
2311104058-development-system [Running] - Oracle VM VirtualBox : 1
File Machine View Devices Help
Activities Terminal Mon 16 Mar, 14:25
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
xsh $ kill 5

Main process recreating shell

-----
XINU
-----

Welcome to Xinu!

xsh $ memstat
Current system memory statistics:
-----
    282563 bytes (0x00044fc3) of Xinu code
    94208 bytes (0x00017000) of allocated stack space
   4062952 bytes (0x003dfee8) of available kernel heap space

Free List:

2311104058-development-system [Running] - Oracle VM VirtualBox : 1
File Machine View Devices Help
Activities Terminal Mon 16 Mar, 14:26
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
xsh $ ping 8.8.8.8
Pinging 8.8.8.8
ping: cannot send ping
xsh $

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:12 | ttyS0
```

The image shows two screenshots of a terminal window titled "xinu@xinu-develop-end: ~/xinu/compile". The terminal is running the xinu operating system.

Top Screenshot: The user enters the command `xsh $ help`. The output lists available shell commands in a grid:

shell commands are:					
argecho	date	ls	ns	sleep	udpserver
arp	devdump	kill	netinfo	tee	uptime
cat	echo	memdump	ping	udp	?
clear	help	memstat	ps	udpecho	

Below the grid, the user enters `xsh $ netinfo`, and the output displays network configuration details:

```
IP address:      192.168.201.115    0xc0a8c973
IP broadcast:    192.168.201.255    0xc0a8c9ff
IP prefix:       192.168.201.0     0xc0a8c900
Address mask:    255.255.255.0     0xffffffff
IP router:       0.0.0.0           0x00000000
NTP time server: 172.21.121.4      0xac157904
MAC unicast:     08:00:27:ae:68:54
MAC broadcast:   ff:ff:ff:ff:ff:ff
```

Bottom Screenshot: The user enters the command `xsh $ ps`. The output shows a table of running processes:

Pid	Name	State	Prio	Ppid	Stack Base	Stack Ptr	Stack Size
0	prnull	ready	0	0	0x005FDFFC	0x00FFFE84	8192
1	ipout	wait	500	0	0x005FBFFC	0x005FBF30	8192
2	netin	wait	500	0	0x005F9FFC	0x005F9ED0	8192
4	Main process	recv	20	3	0x005E7FFC	0x005E7F34	65536
40	shell	recv	20	4	0x005F7FFC	0x005F79AC	4096
58	ps	curr	20	40	0x005F6FFC	0x005F6DE8	8192

Berikut fungsi masing-masing perintah yang muncul pada help.

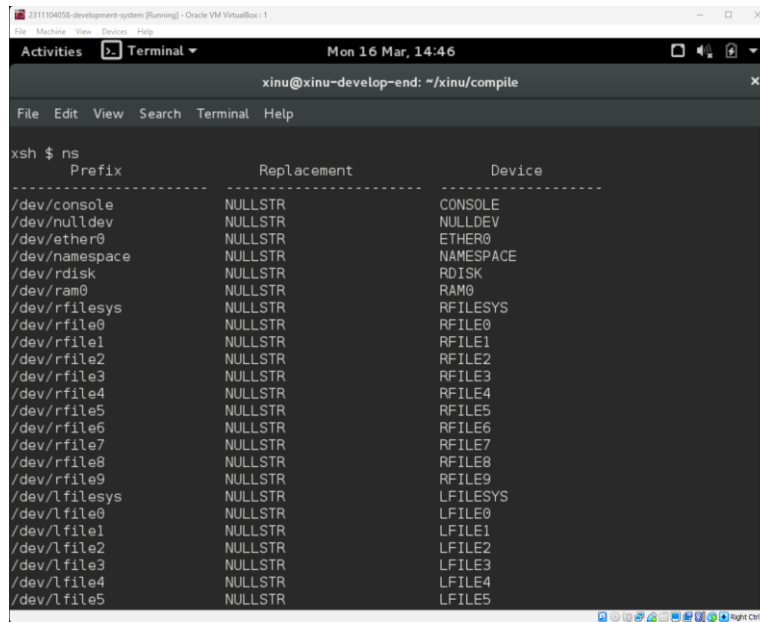
- argecho: Menampilkan kembali argumen yang dimasukkan pada command line.
- arp: Menampilkan ARP cache, yaitu tabel yang berisi pasangan IP address dan MAC address pada jaringan.
- cat: Menampilkan isi file atau membaca input dari keyboard dan menampilkannya kembali ke layar.

- clear: Membersihkan tampilan terminal.
- date: Menampilkan tanggal dan waktu pada sistem.
- devdump: Menampilkan informasi perangkat (device) yang ada pada sistem Xinu.
- echo: Menampilkan teks atau pesan ke terminal.
- help: Menampilkan daftar semua perintah yang tersedia pada shell Xinu.
- kill: Digunakan untuk menghentikan atau mengakhiri suatu proses berdasarkan PID (Process ID).
- ls: Menampilkan daftar file atau direktori.
- memdump: Menampilkan isi memori pada alamat tertentu.
- memstat: Menampilkan statistik penggunaan memori pada sistem Xinu.
- netinfo: Menampilkan informasi jaringan yang digunakan oleh sistem Xinu.
- ns: Digunakan untuk melakukan pencarian name server atau resolusi nama jaringan.
- ping: Digunakan untuk menguji koneksi jaringan dengan mengirim paket ke alamat IP tertentu.
- ps: Menampilkan daftar proses yang sedang berjalan di sistem.
- sleep: Menunda eksekusi proses selama beberapa waktu tertentu.
- tee: Menyalin output ke terminal dan juga ke file secara bersamaan.
- udp: Digunakan untuk melakukan komunikasi jaringan menggunakan protokol UDP.
- udpecho: Program UDP yang mengirim dan menerima pesan echo melalui jaringan.
- udpserver: Menjalankan server UDP pada sistem Xinu.
- uptime: Menampilkan lama waktu sistem telah berjalan sejak terakhir dinyalakan.
- ?: Perintah alternatif untuk menampilkan bantuan (sama seperti help).

2. Pertanyaan

- 22 perintah
- help dan ?
- 192.168.201.1
- arp atau netinfo

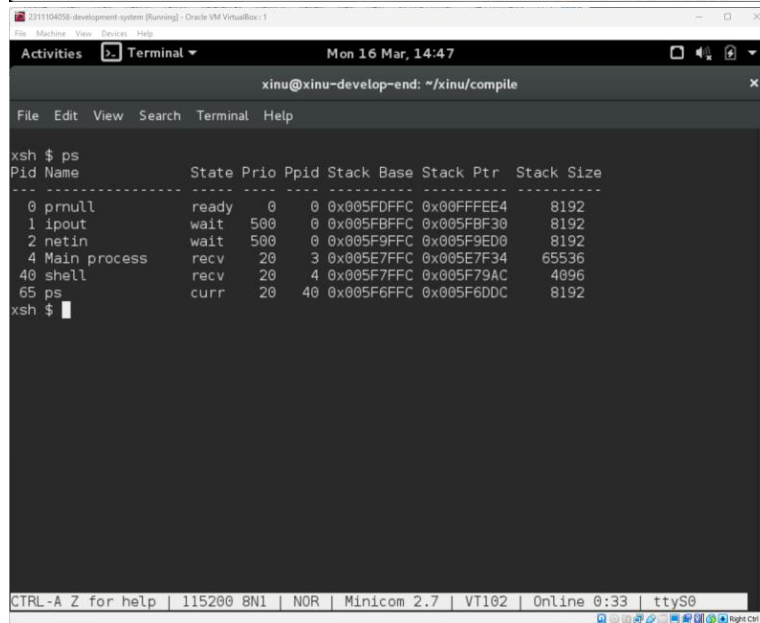
e.



```
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

xsh $ ns
-----
Prefix          Replacement      Device
-----
/dev/console     NULLSTR          CONSOLE
/dev/nulldev     NULLSTR          NULLDEV
/dev/ether0       NULLSTR          ETHER0
/dev/namespace   NULLSTR          NAMESPACE
/dev/rdisk        NULLSTR          RDISK
/dev/ram0         NULLSTR          RAM0
/dev/rfilesys     NULLSTR          RFILESYS
/dev/rfile0       NULLSTR          RFILE0
/dev/rfile1       NULLSTR          RFILE1
/dev/rfile2       NULLSTR          RFILE2
/dev/rfile3       NULLSTR          RFILE3
/dev/rfile4       NULLSTR          RFILE4
/dev/rfile5       NULLSTR          RFILE5
/dev/rfile6       NULLSTR          RFILE6
/dev/rfile7       NULLSTR          RFILE7
/dev/rfile8       NULLSTR          RFILE8
/dev/rfile9       NULLSTR          RFILE9
/dev/lfilesys     NULLSTR          LFILESYS
/dev/lfile0       NULLSTR          LFILE0
/dev/lfile1       NULLSTR          LFILE1
/dev/lfile2       NULLSTR          LFILE2
/dev/lfile3       NULLSTR          LFILE3
/dev/lfile4       NULLSTR          LFILE4
/dev/lfile5       NULLSTR          LFILE5
```

f.



```
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

xsh $ ps
-----
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size
-----
0 prnull          ready  0    0  0x005F0FFC 0x00FFFFE4  8192
1 ipout           wait   500  0  0x005FBFFC 0x005FBF30  8192
2 netin           wait   500  0  0x005F9FFC 0x005F9ED0  8192
4 Main process    recv   20    3  0x005E7FFC 0x005E7F34 65536
40 shell          recv   20    4  0x005F7FFC 0x005F79AC  4096
65 ps             curr   20   40  0x005F6FFC 0x005F6DDC  8192
xsh $
```

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:33 | ttyS0

g.

```
2311104059:development-system [Running] - Oracle VM VirtualBox: 1
Activities Terminal Mon 16 Mar, 14:47
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0 0x005F9FFC 0x005F9E00 8192
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536
40 shell      recv   20   4 0x005F7FFC 0x005F79AC 4096
65 ps         curr   20  40 0x005F6FFC 0x005F6DDC 8192
xsh $
```

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:33 | ttyS0

h.

```
2311104059:development-system [Running] - Oracle VM VirtualBox: 1
Activities Terminal Mon 16 Mar, 14:47
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout       wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0 0x005F9FFC 0x005F9E00 8192
4 Main process recv   20   3 0x005E7FFC 0x005E7F34 65536
40 shell      recv   20   4 0x005F7FFC 0x005F79AC 4096
65 ps         curr   20  40 0x005F6FFC 0x005F6DDC 8192
xsh $
```

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:33 | ttyS0

i.

```
2311104059-development-system (Running) - Oracle VM VirtualBox 1
File Machine View Devices Help
Activities Terminal Mon 16 Mar, 14:47
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0  0x005FDFFC 0x00FFFE4  8192
1 ipout       wait   500  0  0x005FBFFC 0x005FBF30  8192
2 netin       wait   500  0  0x005F9FFC 0x005F9E00  8192
4 Main process  recv   20   3  0x005E7FFC 0x005E7F34 65536
40 shell      recv   20   4  0x005F7FFC 0x005F79AC 4096
65 ps        curr   20  40  0x005F6FFC 0x005F6DDC  8192
xsh $
```

j.

```
2311104059-development-system (Running) - Oracle VM VirtualBox 1
File Machine View Devices Help
Activities Terminal Mon 16 Mar, 14:47
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0  0x005FDFFC 0x00FFFE4  8192
1 ipout       wait   500  0  0x005FBFFC 0x005FBF30  8192
2 netin       wait   500  0  0x005F9FFC 0x005F9E00  8192
4 Main process  recv   20   3  0x005E7FFC 0x005E7F34 65536
40 shell      recv   20   4  0x005F7FFC 0x005F79AC 4096
65 ps        curr   20  40  0x005F6FFC 0x005F6DDC  8192
xsh $
```

k.

```
2311104056-development-system [Running] - Oracle VM VirtualBox 1
Activities Terminal Mon 16 Mar, 14:47
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0  0x005FDFFC 0x00FFFE4 8192
1 ipout       wait   500  0  0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0  0x005F9FFC 0x005F9ED0 8192
4 Main process recv    20  3  0x005E7FFC 0x005E7F34 65536
40 shell      recv    20  4  0x005F7FFC 0x005F79AC 4096
65 ps         curr    20  40 0x005F6FFC 0x005F6DDC 8192
xsh $
```

l.

```
2311104056-development-system [Running] - Oracle VM VirtualBox 1
Activities Terminal Mon 16 Mar, 14:47
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

xsh $ ps
Pid Name      State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull      ready  0    0  0x005FDFFC 0x00FFFE4 8192
1 ipout       wait   500  0  0x005FBFFC 0x005FBF30 8192
2 netin       wait   500  0  0x005F9FFC 0x005F9ED0 8192
4 Main process recv    20  3  0x005E7FFC 0x005E7F34 65536
40 shell      recv    20  4  0x005F7FFC 0x005F79AC 4096
65 ps         curr    20  40 0x005F6FFC 0x005F6DDC 8192
xsh $
```

```
231104058-development-system [Running] - Oracle VM VirtualBox 1
File Machine View Devices Help
Activities Terminal Mon 16 Mar, 14:55
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr Stack Size
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFEF0 8192
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin           wait   500  0 0x005F9FFC 0x005F9E00 8192
4 Main process    recv   20   3 0x005E7FFC 0x005E7F34 65536
74 shell          recv   20   4 0x005F7FFC 0x005F79AC 4096
76 ps            curr   20  74 0x005F6FFC 0x005F6DDC 8192
xsh $ kill 76
kill: cannot kill process 76
xsh $
```

m.

```
231104058-development-system [Running] - Oracle VM VirtualBox 1
File Machine View Devices Help
Activities Terminal Mon 16 Mar, 14:53
xinu@xinu-develop-end: ~/xinu/compile

File Edit View Search Terminal Help

0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192
1 ipout           wait   500  0 0x005FBFFC 0x005FBF30 8192
2 netin           wait   500  0 0x005F9FFC 0x005F9E00 8192
4 Main process    recv   20   3 0x005E7FFC 0x005E7F34 65536
40 shell          recv   20   4 0x005F7FFC 0x005F79AC 4096
72 ps            curr   20  40 0x005F6FFC 0x005F6DDC 8192
xsh $ kill 40

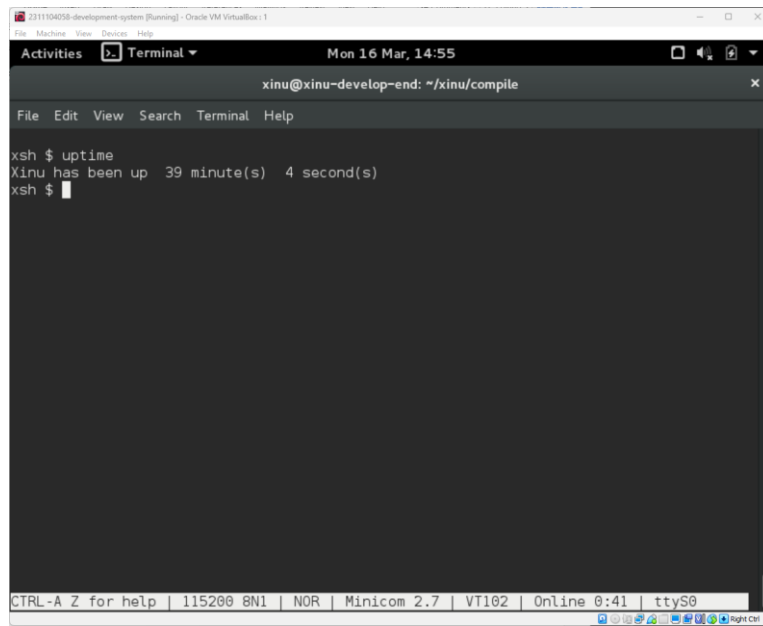
Main process recreating shell

-----
XINU
-----

Welcome to Xinu!

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:39 | ttyS0
```

n.

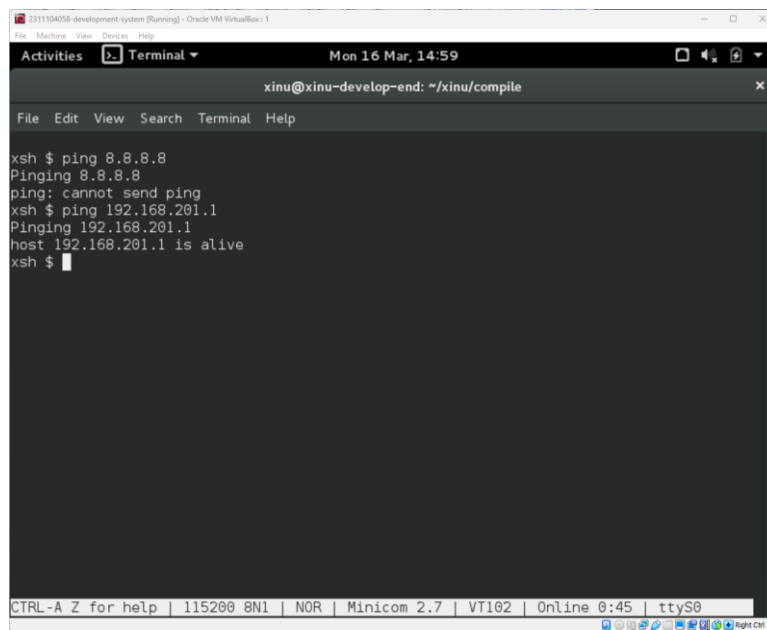


A screenshot of a terminal window titled "xinu@xinu-develop-end: ~/xinu/compile". The terminal shows the command `xsh $ uptime` and its output: `Xinu has been up 39 minute(s) 4 second(s)`. The terminal has a menu bar with "File", "Edit", "View", "Search", "Terminal", and "Help". The status bar at the bottom shows "CTRL-A Z for help", "115200 8N1", "NOR", "Minicom 2.7", "VT102", "Online 0:41", and "ttyS0".

```
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

xsh $ uptime
Xinu has been up 39 minute(s) 4 second(s)
xsh $
```

- o.
- p. clear
- q. memstat



A screenshot of a terminal window titled "xinu@xinu-develop-end: ~/xinu/compile". The terminal shows two ping commands and their outputs. The first command is `xsh $ ping 8.8.8.8`, which outputs `Pinging 8.8.8.8` and `ping: cannot send ping`. The second command is `xsh $ ping 192.168.201.1`, which outputs `Pinging 192.168.201.1` and `host 192.168.201.1 is alive`. The terminal has a menu bar with "File", "Edit", "View", "Search", "Terminal", and "Help". The status bar at the bottom shows "CTRL-A Z for help", "115200 8N1", "NOR", "Minicom 2.7", "VT102", "Online 0:45", and "ttyS0".

```
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

xsh $ ping 8.8.8.8
Pinging 8.8.8.8
ping: cannot send ping
xsh $ ping 192.168.201.1
Pinging 192.168.201.1
host 192.168.201.1 is alive
xsh $
```

- r.

```
231104058-development-system [Running] - Oracle VM VirtualBox 1
File Machine View Devices Help
Activities Terminal Mon 16 Mar, 15:00
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

xsh $ ping 8.8.8.8
Pinging 8.8.8.8
ping: cannot send ping
xsh $ ping 192.168.201.1
Pinging 192.168.201.1
host 192.168.201.1 is alive
xsh $ echo HELLO WORLD
HELLO WORLD
xsh $

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:46 | ttyS0
```

S.

```
231104058-development-system [Running] - Oracle VM VirtualBox 1
File Machine View Devices Help
Activities Terminal Mon 16 Mar, 15:00
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help

xsh $ ping 8.8.8.8
Pinging 8.8.8.8
ping: cannot send ping
xsh $ ping 192.168.201.1
Pinging 192.168.201.1
host 192.168.201.1 is alive
xsh $ echo HELLO WORLD
HELLO WORLD
xsh $

+-----+
| Leave Minicom? |
| Yes No |
+-----+

CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Online 0:46 | ttyS0
```

t.